6 A nichrome wire X of length 45 cm and cross-sectional area $4.7 \times 10^{-7}\,m^2$ is connected into the circuit shown in Fig. 6.1.

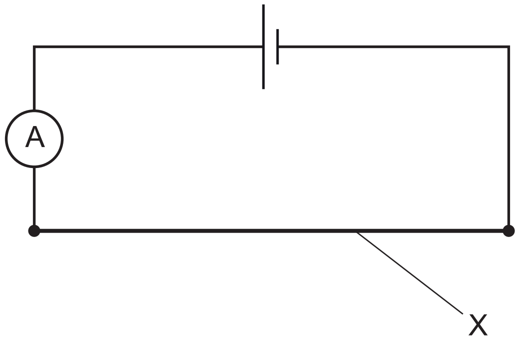

Fig. 6.1

The resistance of X is $1.1\,\Omega$. The cell has electromotive force (e.m.f.) 1.3 V and negligible internal resistance.

(a) **(i)** Calculate the current in X.

current = ... A [1]

(ii) The number density of charge carriers (electrons) in nichrome is $8.5 \times 10^{28}\,m^{-3}$.

Calculate the average drift speed of the charge carriers in X.

average drift speed = ... $m\,s^{-1}$ [2]

(iii) Calculate the resistivity of the nichrome.

resistivity = ... $\Omega\,m$ [3]

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